Python Programming Assignment

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Question 38(a): Vehicle Rental System (10 Marks)

A vehicle rental company stores vehicle details in XML format and rental records in CSV format. Write a Python program to identify vehicles that are currently available for rent and generate an availability report.

Sample Input: vehicles.xml

```
<Vehicles>
    <Vehicle>
        <VehicleID>V101</VehicleID>
        <Type>Sedan</Type>
        <Model>Honda City</Model>
        <Status>Available</Status>
    </Vehicle>
    <Vehicle>
        <VehicleID>V102</VehicleID>
        <Type>SUV</Type>
        <Model>Toyota Fortuner</Model>
        <Status>Rented</Status>
    </Vehicle>
    <Vehicle>
        <VehicleID>V103</VehicleID>
        <Type>Hatchback</Type>
        <Model>Maruti Swift</Model>
        <Status>Available</Status>
    </Vehicle>
    <Vehicle>
        <VehicleID>V104</VehicleID>
        <Type>Sedan</Type>
        <Model>Hyundai Verna</Model>
        <Status>Available</Status>
    </Vehicle>
    <Vehicle>
        <VehicleID>V105</VehicleID>
        <Type>SUV</Type>
        <Model>Mahindra XUV700</Model>
        <Status>Rented</Status>
    </Vehicle>
</Vehicles>
```

Sample Input: rental_records.csv

```
RentalID,VehicleID,CustomerName,RentDate,ReturnDate
R1,V102,Arun Kumar,2026-07-01,
R2,V105,Priya Sharma,2026-07-05,
R3,V101,Karthik Raj,2026-06-20,2026-06-25
R4,V103,Divya Menon,2026-06-15,2026-06-18
```

Python Program

```
import xml.etree.ElementTree as ET
import csv

# Read vehicle details from XML
tree = ET.parse('vehicles.xml')
vehicles = {}
for v in tree.getroot().findall('Vehicle'):
    vid = v.find('VehicleID').text
    vehicles[vid] = {
```

```
        'Type': v.find('Type').text,
        'Model': v.find('Model').text,
        'Status': v.find('Status').text
    }

# Read rental records from CSV, find vehicles not yet returned
rented = set()
with open('rental_records.csv') as f:
    for row in csv.DictReader(f):
        if not row['ReturnDate'].strip():
            rented.add(row['VehicleID'])

# Vehicles available = Status is Available AND not currently rented
available = [(vid, v) for vid, v in vehicles.items()
             if v['Status'] == 'Available' and vid not in rented]

# Generate report
report = ["VEHICLE AVAILABILITY REPORT", f"{'ID':<8}{'Type':<12}{'Model'}"]
for vid, v in available:
    report.append(f"{vid:<8}{v['Type']:<12}{v['Model']}")
report.append(f"\nTotal available: {len(available)}")

report_text = "\n".join(report)
print(report_text)
with open('availability_report_short.txt', 'w') as f:
    f.write(report_text)
```

Sample Output

```
VEHICLE AVAILABILITY REPORT
ID      Type        Model
V101    Sedan       Honda City
V103    Hatchback   Maruti Swift
V104    Sedan       Hyundai Verna

Total available: 3
```

Question 38(b): University Research Project Management (10 Marks)

A university stores faculty details in JSON format and research project information in CSV format. Write a Python program to combine the datasets and generate a report showing the number of research projects handled by each faculty member.

Sample Input: faculty.json

```
[
    {"FacultyID": "F001", "Name": "Dr. Suresh Babu", "Department": "Computer Science"},
    {"FacultyID": "F002", "Name": "Dr. Anitha Rani", "Department": "Electronics"},
    {"FacultyID": "F003", "Name": "Dr. Vishwa T", "Department": "Information Technology"},
    {"FacultyID": "F004", "Name": "Dr. Ramesh Kumar", "Department": "Mechanical"}
]
```

Sample Input: research_projects.csv

```
ProjectID,ProjectTitle,FacultyID,Year
P1,AI in Healthcare,F001,2024
P2,Smart Grid Systems,F002,2023
P3,IoT for Agriculture,F003,2024
P4,Robotics in Manufacturing,F004,2023
P5,Deep Learning for NLP,F001,2025
P6,Renewable Energy Systems,F002,2025
P7,Cloud Security,F003,2025
```

Python Program

```
import json
```

```
import csv

# Read faculty details from JSON
with open('faculty.json') as f:
    faculty = {fac['FacultyID']: fac for fac in json.load(f)}

# Count projects per faculty from CSV
counts = {fid: 0 for fid in faculty}
titles = {fid: [] for fid in faculty}
with open('research_projects.csv') as f:
    for row in csv.DictReader(f):
        fid = row['FacultyID']
        if fid in counts:
            counts[fid] += 1
            titles[fid].append(row['ProjectTitle'])

# Generate report
report = ["FACULTY RESEARCH PROJECT REPORT", f"{'ID':<8}{'Name':<20}{'Dept':<20}{'Projects'}"]
for fid, info in faculty.items():
    report.append(f"{fid:<8}{info['Name']:<20}{info['Department']:<20}{counts[fid]}")

report.append("\nDetails:")
for fid, info in faculty.items():
    report.append(f"{info['Name']}: {', '.join(titles[fid]) or 'None'}")

report_text = "\n".join(report)
print(report_text)
with open('research_report_short.txt', 'w') as f:
    f.write(report_text)
```

Sample Output

```
FACULTY RESEARCH PROJECT REPORT
ID      Name                Dept                Projects
F001    Dr. Suresh Babu     Computer Science    2
F002    Dr. Anitha Rani     Electronics         2
F003    Dr. Vishwa T        Information Technology2
F004    Dr. Ramesh Kumar    Mechanical          1

Details:
Dr. Suresh Babu: AI in Healthcare, Deep Learning for NLP
Dr. Anitha Rani: Smart Grid Systems, Renewable Energy Systems
Dr. Vishwa T: IoT for Agriculture, Cloud Security
Dr. Ramesh Kumar: Robotics in Manufacturing
```